# Robust, Nonparametric, and Shrinkage Estimation

Applied Linear Models I • Lecture 15 of 16

Applied Linear Models I

# Today: when OLS is not enough

- **Robust** regression – resist outliers (M-estimation).
- **Breakdown point** – mean vs median.
- **Nonparametric** regression – loess.
- **Shrinkage** – James–Stein and Stein's paradox.

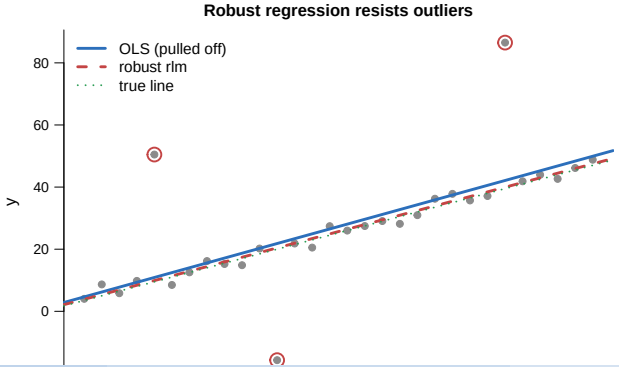
Theme: **unbiasedness and least squares are not sacred.**

# Robust regression: M-estimation

OLS squares residuals $\Rightarrow$ one outlier dominates. Replace the square:

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} \sum_i \rho\left(\frac{y_i - \mathbf{x}_i^\top \boldsymbol{\beta}}{\hat{\sigma}}\right),$$

Huber $\rho$: quadratic near 0, linear in the tails (**bounded influence**).



# Robust weights and the breakdown point

**IRLS down-weights** (not deletes): outlier weights $\approx 0.06$ vs $\approx 1$ for clean points.

**Breakdown point** – fraction of bad data tolerated:

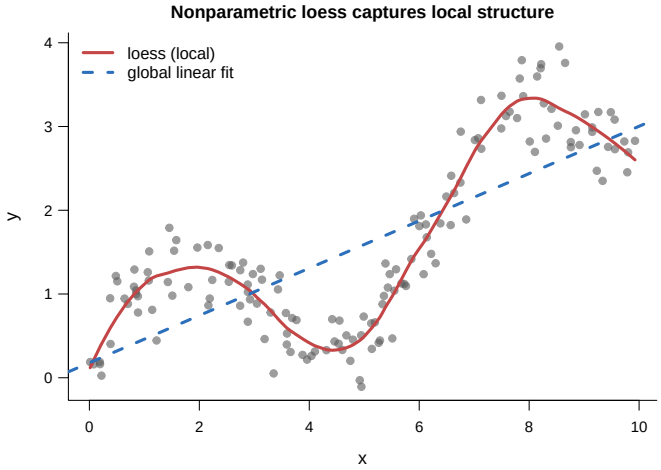
- mean: 0% (one point breaks it)
- median: **50%** (the best possible)

**One huge outlier among 99 points**

mean: $10.0 \to 10{,}010$ median: $10.00 \to 10.02$.

# Nonparametric regression: loess

Fit a local low-degree polynomial in a moving neighborhood; tune the **span** (bias vs variance).



# Stein's paradox

Estimate $p \geq 3$ means from one noisy observation each. The MLE (use each as-is) is **inadmissible** – beaten by James–Stein:

$$\hat{\boldsymbol{\theta}}_{\mathrm{JS}} = \left(1 - \frac{p-2}{\|\mathbf{z}\|^2}\right)\mathbf{z}.$$

**10 means, 4000 replications**

Total MSE: MLE $= 10.08$ vs James–Stein $= 8.11$.

Shrinking each estimate using *unrelated* coordinates lowers *total* risk.

# Shrinkage = ridge = the central trade

- James–Stein shrinks means; **ridge** (L10) shrinks coefficients.
- Both are **deliberately biased** and lower MSE.
- Gauss–Markov finds the best *unbiased* estimator – dropping that constraint can do strictly better.

**The course's key lesson**

**Accuracy, not unbiasedness, is the goal.**

# Summary

- M-estimators down-weight outliers via IRLS; OLS does not resist them.
- Breakdown: mean 0%, median 50%; use high-breakdown methods if needed.
- loess fits local polynomials (span tunes bias–variance).
- James–Stein dominates the MLE for $p \geq 3$; shrinkage = ridge.

**Next**

Lecture 16: generalized linear models (then the final).